Wildfire Smoke & PM2.5 Forecasting

CS2: Transformer-Based Air Quality Prediction

Why PM2.5 Forecasting Matters

Motivation

The asthma rate in the BC area has nearly doubled, rising from 5.25% to 9.88%.

Primary Stakeholder

Environmental Health Director — responsible for timely, defensible air-quality advisories used across provincial partners.

CS1 Limitations and CS2 Objectives

CS1 Limitations

- Relied on historical PM2.5 patterns; using too few features risks over-reliance on a single factor.
- Lacked short- and long-term memory, which is problematic for time-series data.
- The model was too conservative, causing it to miss rapid, non-linear spikes (notably from wildfire smoke).

CS2 Objectives

- Expand the feature set to capture environmental complexity.
- Incorporate memory into the model to better handle time-series dynamics.
- Improve the model's sensitivity so it is less conservative during extreme events.

Data Pipeline and Feature Engineering

Two data sources (2022–2025):

- **PM2.5** — CS1 pipeline (BC AQMS, zone-median, hourly, PST $\rightarrow$ UTC $+8$h)
- **Wildfire** — CWFIS satellite hotspots (MODIS/VIIRS, UTC)

Key design decisions:

- Zone **polygon boundary** distance, not centroid
- **300 km** regional radius — cross-zone smoke transport
- IDW weighting $(1/d^2)$ — closer fires weighted more
- Non-fire hours **zero-filled**, not missing

Wildfire signals:

- `fire_count`, `frp_sum` (local & regional)
- `hfi_weighted`, `fwi_mean`

Engineered features:

Feature	Values
PM2.5 lags	1h, 6h, 24h, 48h
Rolling	24h mean/max, 7d mean
Fire rolling	frp 24h/72h sum
Calendar	hour, dow, month, season

Data Structure and Modeling Assumptions

Data Structure:

- 70,128 rows × 11 features, 2 zones, hourly UTC
- Sliding window: $X = (N,\ 72,\ 11)$, $y =$ PM2.5 at $t + 3$
- Temporal split — Train: 2022–2024 Test: 2025

Data Augmentation:

- **PCHIP interpolation** — upsample hourly $\rightarrow$ 30-min
- Doubles training sequences; smoother PM2.5 transitions
- *Assumption: more data improves model generalization*

Data Assumptions:

1. **Wildfire drives PM2.5** — spikes in BC air quality are primarily caused by wildfire smoke
2. **300 km influence radius** — fires within 300 km of a zone boundary meaningfully affect local PM2.5
3. **Zone independence** — Lower Fraser Valley and Southern Interior modeled separately; no cross-zone features

Model Architecture and Design Choice

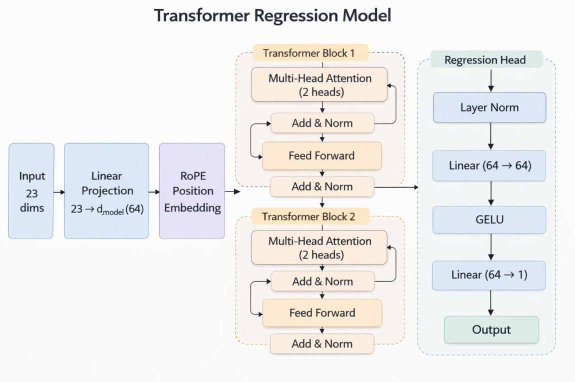
Transformer Model Structure

Architecture

- Transformer encoder
- Sequence input: **96 hours**
- d_model = 64
- 2 attention heads
- 2 Transformer layers
- dropout = 0.1

Total parameters size: 0.106M

Total parameters intentionally **kept small** to reduce overfitting.



Why Transformer? Model Selection and Trade-offs

From Baseline GAM to Transformer

Model	Improves	Still limited by
GAM (baseline)	Nonlinear structure	Manual features, weak sequence modeling
LightGBM	Identify useful predictors for nonlinear tabular learning	Limited time-series handling
LSTM + Autoencoder	Sequence learning + feature compression	Slow sequential training

Therefore, we chose Transformer: Captures temporal dependence more flexibly and supports parallel training.

Why Transformer

- **Sequence-aware** models historical PM2.5 patterns directly
- **Long-range dependence** can connect distant time steps more effectively
- **Flexible representation** captures nonlinear temporal structure
- **Parallel training** more scalable than recurrent models

Trade-offs

- **Higher compute cost** larger and more complex than GAM
- **More tuning required** architecture and optimization settings matter
- **Lower interpretability** harder to explain than smooth GAM effects
- **Overfitting risk** requires careful validation and regularization

Why Not More Complex Alternatives?

We experimented with **more complex architectures** inspired by LLaMA:

- SwiGLU feed-forward layers
- Larger hidden dimensions
- Deeper Transformer blocks

However:

- Dataset size relatively small: 26,245 data entries (12.05 MB) per zone
- Models trained **separately per zone**
- Complex variants did **not improve stability or performance**

Final decision:

Use a simpler Transformer block with strong regularization.

Model Fitting and Tuning

Training Setup

Optimizer: **AdamW**

Loss: **MSELoss**

Training parameters:

- Batch size: 256
- Epochs: 10
- Learning rate: 5e-4 $\rightarrow$ 5e-6 (cosine decay)
- Weight decay: 0.05

Hyperparameter search was performed using **manual exploration**.

Feature and Window Experiments

Feature set includes:

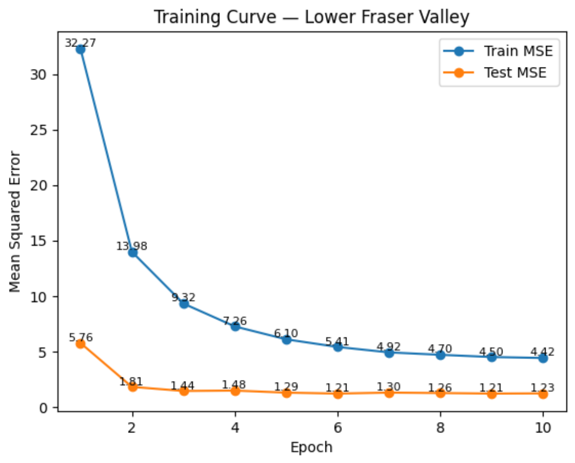
- PM2.5 lags and rolling statistics
- Fire activity indicators (FRP, fire counts)
- Temporal encoding (hour, weekday, month)

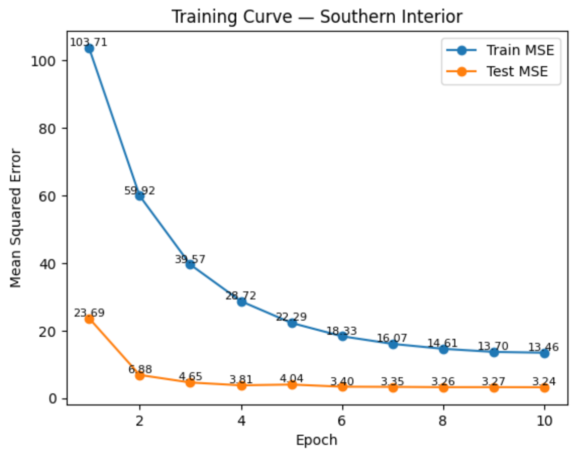
Window length experiment:

12h, 24h, 48h, 72h, **96h**, 168h

Best performance observed at **96 hours**.

Training Dynamics





Training monitored using **train and test loss curves** across 10 epochs.

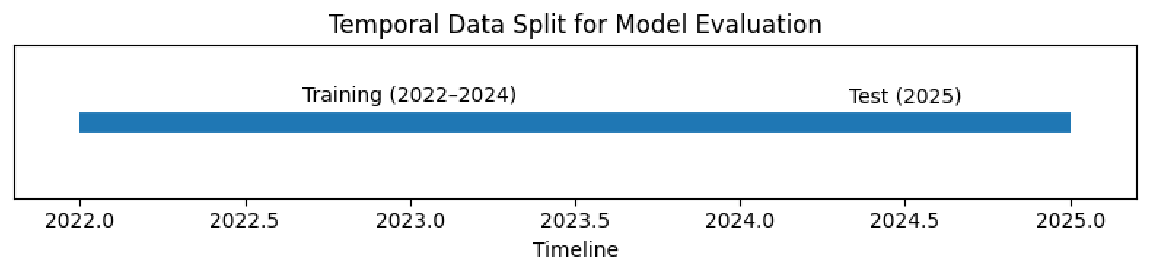
Final Model Performance

Zone	RMSE	MAE	R^2	N_test
Lower Fraser Valley	1.108	0.659	0.843	8710
Southern Interior	1.801	1.230	0.863	8710

Validation Strategy and Leakage Prevention

Time-Series Evaluation

Random data splits are **inappropriate for time-series data**, because they allow future observations to influence training.
Instead we use **temporal splitting**.



Train: **2022–2024**
Test: **2025**

Leakage Prevention

Several safeguards were applied:

- StandardScaler **fit only on training data**
- Test data transformed using training statistics
- Sequence windows constructed **after dataset split**
- No window crosses train–test boundaries

Zone-Specific Modeling

Models trained **independently** for each region:

- Lower Fraser Valley
- Southern Interior

This avoids mixing **regional wildfire patterns**.

Model predictions follow **PM2.5 trends in the unseen test year**, demonstrating generalization to new wildfire seasons.

Main Performance Comparison

Zone	Model	Precision@25	Recall@25	F1@25	F2@25
Lower Fraser Valley	Baseline	0.500	0.538	0.518	
	Transformer	0.538	0.538	0.538	0.538
Southern Interior	Baseline	0.945	0.896	0.920	
	Transformer	0.945	0.888	0.916	0.899

- Model performance differed sharply by air zone.
- Southern Interior performed strongly: Precision 0.945, Recall 0.888, F1 0.916, F2 0.899.
- Lower Fraser Valley performed much worse: all key metrics were 0.538.
- Overall, the model appears reliable in the Southern Interior but much less dependable in the Lower Fraser Valley.

Diagnostics and Air-Zone Comparison

- Diagnostics show that performance is highly region-dependent.
- The Southern Interior appears to have clearer or more stable event patterns, which the model captured well.
- The Lower Fraser Valley appears to be a more difficult forecasting environment, leading to weaker warning performance.
- This comparison shows why zone-level evaluation matters: overall results alone would hide important regional differences.

Limitations, Uncertainty, and Final Takeaways

- Results should be interpreted with caution because performance is not consistent across air zones.
- The weaker Lower Fraser Valley results indicate greater uncertainty and a higher risk of unreliable warnings.
- The approach shows clear promise, especially for the Southern Interior.
- Main takeaway: the model has value, but further refinement and zone-specific tuning are needed before broader operational use.